

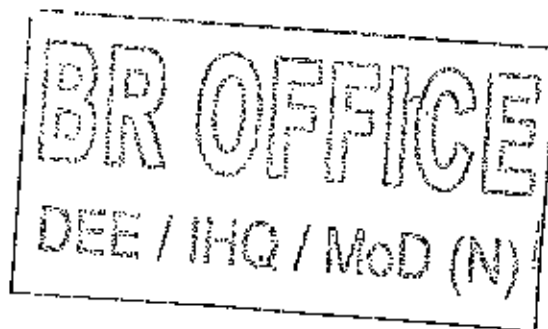
STATEMENT OF TECHNICAL REQUIREMENTS

FOR FITMENT OF

DIFFERENTIAL GLOBAL POSITIONING SYSTEM

EED-56-02

DEC 2006



MATERIAL BRANCH
INTEGRATED HEADQUARTERS
MoD (NAVY)
DIRECTORATE OF ELECTRICAL ENGINEERING
'D' WING, SENABHAWAN
NEW DELHI - 110 011

HISTORY RECORD**I. AMENDMENT RECORD**

Amendment No.	Date	Text Affected	Signature/Date
		NIL	
		NIL	

II. REVISION NOTE

Revision No.	Date	Text Affected	Signature/Date
		NIL	
		NIL	

III. SUPPLEMENT RECORD

Supplement No.	Date	Text Affected	Signature/Date
		NIL	
		NIL	

This standard supersedes the following:-

EED No.	Date	Signature/Date	Remarks
		NIL	
		NIL	

CONTENTS

SI No	Description	Page No
1	Introduction	4 - 5
2	Environmental, EMI – EMC	5 - 7
3	Onboard And B&D Spare	8 - 9
4	Miscellaneous Requirements	11-13

List of Appendices

SI No	Description	Page No
1	Appendix A - Environmental Stress Screening Specifications	14
2	Appendix B - IN Maintenance Philosophy	15

DIFFERENTIAL GLOBAL POSITIONING SYSTEM

1. **INTRODUCTION.** DGPS Navigation uses satellite ranging to dateline your position in relation to a set of satellites orbiting the earth. The DGPS constellation made up of 24 satellites which continuously send radio signals containing precise position and time information for each satellite base to earth, by knowing the position of any 3 or 4 of these satellites. The Differential Global Positioning System is an enhancement to Global Positioning System that uses a network of fixed ground based reference stations to broadcast the difference between the positions indicated by the satellite systems and the known fixed positions. These stations broadcast the difference between the measured satellite pseudo ranges and actual (internally computed) pseudo ranges and receiver stations correct the pseudo ranges by the same amount.

1.1 **Performance.** System should have state of the art multi track TM receiver tracks up to eight satellites while providing high receiver sensitivity, fast first fix and continuous navigation updates.

1.2 **Probability.** System should have the built in simulator allows you to plan your next trip or sharpen one's navigation skills at ship.

1.3 **Low power consumption.** System should have Battery saver operation, which draws less than 250 millions of current, provides 14 hours of continuous operation with six 44 size alkaline batteries. The quick fix TM mode provides on demand position fixes in as little as 15 seconds and over 100 hours of battery life at four fixes per hour.

1.4 **Multi-purpose Alarm.** System should have Dual alarm clock and timers allow the DGPS 100 to watch to clock for you. Arrival anchor drag, CDI and hazard alerts help you safety navigate craft.

1.5 **Offset Navigation.** System should have the offset navigation treat allow you to transfer your route a fixed distance left or right from a defined course.

1.6 **Interfaces.** The system should be integrated with IBS, Auto Pilot, RLGs, ECDIS and AISDN. The DGPS display should be installed on the Bridge.

1.7 **Physical**

Case -	Splash and corrosion proof
Size -	6.25"W X 3.95"D X 2"H (159mm) X 100 mm 51 mm
Weight -	25 ounces - 0.71 Kg with bat pack. 28 ounces - 0.80 Kg with rechargeable battery pack
Power -	-10 - 33 UDC with power cable 115 V or 230 V AC with battery charger.

Current 245 milliamps in normal mode using battery (44.5 milliamps with full back lighting) 155 milliamps in battery saver mode when using battery saver mode when using battery (355 milliamps with full backlighting) 145 milliamps when using 12 VDC external power (265 milliamps with full backlighting).

1.8 **Failure Warning and Status Indication.** The system should in addition to normal GPS operation related failures should be capable of proving:-

(a) Status of indication of:

(i) Receipt of DGPS Signals.

(ii) whether DGPS corrections are being applied to the indicated ship's position.

(b) DGPS integrity status and alarm.

(c) DGPS text message display.

1.8.1 **Frequency.** The equipment should operate in dual frequency capable and should be capable of automatic selection of frequency.

1.9 **ENVIRONMENTAL:-** The system should be able to withstand the following temperature and other requirements.

-15° C to + 70° C (+5 F to 158 F) operating
-40° C to + 70° C (-40 F to 158 F) Storage

Humidity	-	95% non condensing
Receiver	-	Multitask TM Tracking upto 8 satellites.
AC Allision time	-	2 minutes 2D
Typical	-	2.5 minutes 3D 15 Seconds quick fix
Update Rate	-	Once per second, continuously
Accuracy	-	Both the static and the dynamic accuracies Should be 10m or better (95%)
Dynamics	-	695 Knots velocity, 3g
Optional accessories	-	The following optional accessories are available for your specific needs:-

- (a) Portable antenna
- (b) Magnetic Mount antenna
- (c) Marine/RV remote antenna kit
- (d) Marine/Mobile mounting kit
- (e) Personal computers kit
- (f) Carrying case
- (g) AA size battery pack.

2. **EMI – EMC.** The entire system should have total electro-magnetic compatibility with all shipboard equipment. The EMI/EMC levels should conform to MIL-STD 461C/D and be subjected to the following tests: -

S No	Test	Specification
(a)	Conducted Emission	CE01
(b)	Conducted Emission	CE03
(c)	Conducted Susceptibility	CS01
(d)	Conducted Susceptibility	CS02

(e)	Conducted Susceptibility	CS06
(f)	Radiated Emission	RE01
(g)	Radiated Emission	RE02
(h)	Radiated Susceptibility	RS01
(i)	Radiated Susceptibility	RS02
(k)	Radiated Susceptibility	RS03

Environmental Conditions.

3. The following Environmental Conditions are specified for the above deck equipment.

3.1 Temperature:

15° C to + 70° C (+5 F to 158 F) operating
-40° C to + 70° C (-40 F to 158 F) Storage

- 3.2 **Humidity.** Relative humidity conditions of up to 95%, including condensation in the form of water and frost.

- 3.3 **Rain.** The system should be capable of withstanding in operation and in non-operation conditions, exposed to a rain showers in accordance with MIL-STD 108.

- 3.4 **Sand and Fog.** The system should be capable of withstanding in operation and in non-operation conditions exposed to sand and fog in accordance with MIL-STD 810E, method 509.3, procedure I.

- 3.5 **Ice.** The system should be capable of withstanding external climatic conditions of ice on surface without any damage to the surface.

- 3.6 **Wind.** The system should be withstand wind speeds of 70 km/hour maximum without decreasing of performance.

- 3.7 **Fungus.** The system should be withstands, in both operating and non-operating modes, exposure to fungus growth as encountered in tropical climates.

- 3.8 **Salt Atmosphere.** The system should be withstands, in both operating and non-operating modes, exposure to salt-sea atmosphere.

- 3.9 **Lightning Protection.** All external I/O and power lines includes lightning protection circuitry, according to requirements of IEC-1000-4-5, level 4, for shielded lines - potential difference.

- 3.10 **Altitude.** The system should be suitable for air transportation without suffering damage due to the altitude.

- 3.11 **Vibration.** The system should be capable of withstanding vibration in accordance with MIL-STD 810E, method 514.4 as defined in Figure 514, 4-15.

- 3.12 **Shock.** The system should be capable of withstanding vibration requirements in accordance with MIL-STD 810E, method 516.4, and procedures I-3.3 and II-3.1.

- 3.13 **Solar Radiation.** All the external equipment's doing not suffer any damage and remains operative during the solar radiation.

4. **Environmental Specifications** – Specifications for the below deck equipment is as follows:-

#	TEST	Standard and Method
(a)	Vibration Test	IAW JSS-55555: 2000, Test No. 28, 5-33 Hz, AMP $\pm$ 0.125 mm, constant displacement
(b)	High Temp	IAW JSS-55555: 2000, Test No. 17, Operating at + 50 deg C Storage at + 70 deg C Procedure 6, Test condition 'K'
(c)	Damp Heat	IAW JSS-55555: 2000, Test No.10, Operating Temp. 40 deg C, RH 95 %
(d)	Low Temp	IAW JSS-55555: 2000, Test No. 20, Operating Temp. 0 deg C
(e)	Drip Proof	IAW JSS-55555: 2000, Test No. 11,
(f)	Tropical Exposure	IAW JSS-55555: 2000, Test No. 27, Temp variation from 20 deg C to 35 deg C, RH 95 %
(g)	Mould Growth (Representative samples)	IAW MIL-STD-810 E, Method 508.4, Sec. II
(h)	Corrosion (Salt) (Representative samples)	IAW MIL-STD-810 E, Method 509.3, Proc. I
(j)	Bump Test	IAW JSS-55555: 2000, Test No. 5, 1000 bumps, vertical axis
(k)	Shock Test	IAW JSS-55555: 2000, Test No. 24, 40g, 11 ms, Vertical axis IAW JSS-55555: 2000, Test No. 28,
(l)	Toppling	IAW JSS-55555: 2000, Test No. 26,

5. **Environmental Stress Screening (ESS) Test.** ESS test for electronic components/sub systems of Dynamic Global Positioning System, is required to be carried out for 100% components/parts/assemblies. The severities of ESS test are placed at Appendix B.

6. **Software Standards.** Software standards if any inbuilt in the system should be as per MIL STD 2167A/498. The following software documents are to be supplied

#	Document Description
(a)	SDP – Software Development Plan
(b)	IDD – Interface Design Document
(c)	STPD – Software Test Plan and Description

(d)	STR – Software Test Results
(e)	Per CSCI: SRS – Software Requirements Specifications
(f)	Per CSCI: SDD – Software Design Document
(g)	Per CSCI: SVD – Software Version Description

7. **MTBF/MTTR.** The MTBF should be more than 3000 running hours. The MTTR should not exceed 30 minutes for the electronics and one hour for the other components.

8. **Supplier Scope Of System Supply, Installation And Commissioning.**

8.1 The supplier's scope of system supply shall cover the supply of equipment, Installation Material, Spares, Drawings and Documentation, installation supervision, setting to work, testing, on board commissioning, harbour and sea trials of Dynamic Global Positioning System. The supplier is to list down in detail, the deliverables to be provided to the customer at various stages of supply as mutually agreed upon. The complete scope of supply of the system by supplier per ship shall include as given in succeeding paragraphs.

8.2 **Installation Material.** One set of installation material that includes inter-cabinet cables, connectors, shock mounts, special fasteners, special tools, special fittings etc. shall be supplied by the supplier.

8.3 **Testing and Tuning Spares.** The supplier shall recommend one set of Testing and Tuning Spares required for STW, HATS/SATS and commissioning of the system.

8.4 **Special Tools and Test Equipment.** The OEM is to provide details of following test equipment that would be required taking into account IN's maintenance philosophy given at Appendix 'C'.

(a) **Onboard Test Equipment.** This should be adequate to meet the requirement of all repairs/maintenance expected to be carried out onboard ship by the ships staff.

(b) **Special test equipment.** Special-to-type test equipment shall be made available in the shore base for specific tests/checks on the equipment.

(c) **Dockyard Test Equipment.** This should cater for the 3rd and 4th level repair/maintenance.

9. **On Board Spares (OBS).** The supplier shall prepare the list of OBS (including onboard maintenance tools) based on the Reliability and Maintainability data and taking into account IN's maintenance philosophy given at Appendix 'C', and shall forward to IHQ MOD (Navy) for vetting. The list of recommended spares should be supplied at least two months prior to the delivery of the Dynamic Global Positioning System. The manufacturer should clearly indicate the basis for ranging and scaling of spares. This should include the list of spare parts, tools and accessories, which must be carried on the ship, for preventive maintenance, trouble shooting and quick repairs to ensure no more than 30 minutes of 'Down Time' for the system at any given instance. These should cater to first and second level maintenance and should be stocked to take care of one year's requirement. One set of OBS is to be supplied with the system. The finalised / approved list of spares is to be provided in INCAT (Indian Naval Catalogue of Inventory) compatible format in electronic media for ILMS (Integrated Logistics Management system) of Indian Navy, for the management of spares. The supplier shall provide both hard copy as well as soft copy.

10. **Five Year Base and Depot Spares.** The OEM should forward recommended list of B&D spares for Launcher and Fire Control System for vetting of IHQ MOD (Navy). B & D Spares should consist of spare parts and modules required to replenish depot holdings for a period of five years. These should also contain complete subsystems and assembly spares for the major refit (and thorough checking thereafter) of the system. It should also include exclusive Depot Spares comprising components for carrying out spares of PCBs up to component level, using workshop level automatic test facilities available with IN. IN's maintenance philosophy is given at Appendix 'C'. The manufacturer should clearly indicate the basis for ranging and scaling of spares. The list of spares should be prepared by the manufacturer and needs to be supplied at least two months prior to the delivery of the Dynamic Global Positioning System. The supplier shall also provide finalised / approved list of B&D spares in INCAT (Indian Naval Catalogue of Inventory) databank compatible format in electronic media. Order for B&D spares would be placed separately.

11. **Drawings and Documentation.**

(a) Firm is required to submit one hard copy and one soft copy of the draft system documents including software documents to IHQ MOD (Navy) for vetting at least 3 months prior to conduct of FATs of the system.

(b) One set of finalised/approved documents should be supplied with each equipment and an additional **6 sets** (hard copy) and one soft copy on CD ROM of documents for each ship set should be supplied free of cost with the order.

(c) Supply of documents will be part of the order and should be costed in the order.

(d) Documents in CD-ROM are to be as an Interactive Electronic Technical Manual (IETM) as per Indian Navy Specifications EED-P-23.

(e) The format and content of documentation should be as per specifications JSS-0251-01-2002(Rev) and EED-S-048. The vetting and approval of the documents would be done by the AHSP and IHQ, MoD(N).

(f) Software documents should be as per IEEE 12207 standard.

(g) The documentation and drawings shall contain complete information for installation, operation, inspection, maintenance, repair and overhaul, testing and trials and should include the following –

#	Document Description
(i)	Technical and Operating Instruction Manual
(ii)	On Board Maintenance Manual
(iii)	Depot Maintenance Manual
(iv)	Installation and Testing Manual
(v)	Parts and Tools catalogue
(vi)	Installation Drawings
(vii)	Acceptance Test Procedure (ATP) and Test Report (TR)
(viii)	HAT & SAT schedule

12. **Binding Data.** Three hard copies & 2 sets on CD ROM of the following binding drawings/ documents are to be supplied by the OEM within 3 weeks of placement of order: -

- (a) Block diagram of the system
- (b) Installation documents covering detailed procedure for installation with sequence of activities.
- (c) Installation drawings indicating overall dimensions, C.G., weight, maintenance envelope etc of each unit.
- (d) Recommended arrangement of devices in nominated compartments.
- (e) Inter unit-cabling diagram with cables specifications.
- (f) Cable connection Schedule.
- (g) Power supply scheme for the system.
- (h) Heat Dissipation of individual units in compartment & in close loop ventilation system, as required for the system.
- (i) Parts identification list, Indicating part no, qty, maker's name, Specification etc.
- (j) Requirements for support systems such as fire fighting, communication, lighting etc.
- (k) Cable length limitation etc, if any.
- (l) Detailed foundation drawing including bolting plan.

13. **Installation.** Onboard installation of the system will be carried out by the Shipbuilder. The assistance and responsibilities of the OEM include the followings: -

- (a) Coordination with the Shipbuilder for the installation work.
- (b) Providing instructions to the Shipbuilder personnel for installation and mounting of launchers and other units of the system.
- (c) Providing instructions to the Shipbuilder personnel for installation and mounting of the cable connectors of the system. Cable laying will be done by the shipbuilder before start of the installation activities.
- (d) Checking of the cable connections (cold wire test).

14. **Setting-to-Work.** The supplier shall carry out: -

- (a) Setting-to-work.
- (b) Testing and Tuning of the system.

15. **FATs and HATs/SATs.** The supplier to offer the system for FATs, HATs and SATs (including ATFs) as per NHQ approved schedule.

16 **Factory acceptance trials (FATs).** The supplier shall prepare a Factory Acceptance Trials (FATs) schedule. Personnel from Indian Navy will be present for such trials. The schedule should state how the supplier would demonstrate that the delivered system will meet the functional and performance requirements indicated as per Indian Navy's SOTR. The factory acceptance trials procedure shall comprise of

- (a) Functional Tests
- (b) Verification of design, especially firmware and software
- (c) Test equipment used, calibration requirements
- (d) Pass/Fail criteria
- (e) Expected duration and time phasing

16.1 The FATs schedule after being formulated by the supplier should be forwarded to IHQ, MoD(N) for vetting and approval.

16.2 All EMI/EMC tests at para 2.7 need to be completed prior to FATs.

16.3 **Harbour Acceptance Trials (HATs) & Sea Acceptance Trials (SATs).** The supplier is required to provide necessary representative(s) to assist during the following phases:

- (a) HATs (procedure to be approved by the Indian Navy)
- (b) SATs (procedure to be approved by the Indian Navy)

17. **System Performance Responsibility.** In case of any irregularities in the operation/performance of the system or non conformance to specified parameters observed on integration with ships system, the supplier is bound to rectify the defect. The supplier shall ensure complete responsibility of satisfactory operation of the system on board

18. **Interface and Integration.** The supplier is responsible for the interfacing and integration of DGPS with ships onboard systems in accordance with the Interconnectivity matrix promulgated by IHQ MOD (Navy). Interface protocols for interfacing of these systems will be mutually finalised between the manufacturer and Navy. The system should be integrated with IBS, Auto Pilot, RLGs, ECDIS and AISDN. The DGPS display should be installed on the Bridge.

19. **Miscellaneous Requirements**

19.1 **Reliability & Maintainability.** The requirements of reliability and maintainability as described in the Document DOS (W) -S-03/95, must be complied by Dynamic Global Positioning System System.

19.2 **Quality Assurance.** The MQAP, with details of the quality assurance/control programmes proposed to be adopted for the Dynamic Global Positioning System system, should be laid down, along with the guidelines to be followed. These checks would be carried out by the Authorized Inspection Agency, as appointed by the Indian Navy. The IN would carry out the software audit.

19.3 Details of test equipment, test methods, preliminary qualification tests, etc is to be indicated. Shop floor tests prior to dispatch by supplier should be conducted.

19.4 Quality assurance plan proposed and any subsequent amendment shall be discussed and agreed upon between IHQ MoD (Navy) and the manufacturer prior to placement of the order.

19.5 Pre and Post Electrical performance checks during Environmental test of panels need to be witnessed by QA agency. Accordingly, the system supplier should plan his schedule of offering of the components/subsystems and final assembly and the same may be intimated to the QA agency.

19.6 **Product Support.** The supplier should undertake to ensure guaranteed and continuous product support for a period of 15 years from the date of supply of the first system and associated equipment, by way of the following: -

- (a) Supply of spare parts and materials
- (b) Offering of alternative solutions in the event of obsolescence of the components/technologies, including those bought out or subcontracted.
- (c) Continuous up-gradation programme.
- (d) Continuous supply of amendments to the documentation.

19.7 **Repair Facilities.** The manufacturer, along with the details of manpower required, should project facilities, which are required in the Depot.

19.8 **Hardware Warranty.** The warranty period of of Dynamic Global Positioning System system should be for 12 months form the date of completion of SATs.

20. **Training.** The Training program objective is to enable the End User Personal to operate, maintain, and install the Dynamic Global Positioning System system. The Training package is offered as follows (duration in working days).

#	<u>Course Description</u>	<u>No of Participants</u>	<u>Duration (Days)</u>
(a)	Operators(for each ship)	10	5
(b)	On Board Level Maintenance(for each ship)	10	5
(c)	Dockyard Level Maintenance(for each set of test equipment supplied to dockyard)	10	10
(d)	Optional course for Data Demodulator Operators	4	5

21. **Maintenance.** The manufacturer should provide recommended maintenance schedules for preventive and corrective maintenance of the system.

22. **Inspection & Testing.**

(a) **Inspection.**

- (i) Inspection Authority : CQAE/DQAN
- (ii) Receipt Inspection : CSL/Warship Overseeing Team (K)
- (iii) Installation : Warship Overseeing Team (K)
- (iv) Ship trials : WATT (MB)/(V)

(b) The product offered by the manufacturers should conform to standard engineering practices. The system will be subjected to stage inspection and final test

and trials by the Naval Inspection Agencies as mutually agreed with the manufacturer. Any deviation from the mentioned specifications will be brought to the notice of IHQ-Mod (Navy).

(c) The system shall be designed and guaranteed to withstand the shock standards as per JSS 55555 while ensuring system integrity is not affected. There must be no significant effect on performance of essential components that may affect overall ship safety or combat capability and in no way must they be hazardous or dangerous to personnel through failure of system components arising from shock. The manufacturer shall repair the equipment in the event the equipment is damaged as a result of the shock test not exceeding the calculated shock impact loads.

23. **Shipping.** All equipment shall be adequately packed and protected with supports to ensure adequate protection during all modes of transportation. Each unit within a package/container shall be clearly marked for identification. The container shall clearly indicate the item description with caution marks, quantity, weight, size etc. A separate document giving details and instructions for storage, preservations, handling and transportation after delivery is to be supplied. The supplier should indicate the delivery schedule, transport, packing, preservation, insurance etc.

24. **Authority Holding Sealed Particulars (AHSP).** The Indian Navy would be the AHSP for all the sealed particulars of **Differential Global Positioning System** and associated sub systems/equipment.

25. **Preservation, Packaging and Shipping:** The stores (including OBS and B&D Spares) shall be supplied in long-term preserved condition that is suitable for storage under tropical high humidity conditions for a period of 24 months.

(Refers to Para 2.10)

ENVIRONMENTAL STRESS SCREENING

S.No	Test	Specifications	Remarks
1	Thermal shock Thermal Cycling	-40° C/0° C (for commercial grade components) - 90 Min +70° C - 90 Min Change over to take place within 5 min No of cycle - 06 OR 030 to + 55° C change @ 1° C/min 3 cycle dwell time up to 20 kg - 2 hrs 20-40 kg - 3 hrs above 40 kg - 4 hrs	100% on all PCB Sub-unit/equipment with power on
2	Vibration Test	Random vibration freq 20-2000 Hz Power density 02g2 Hz - 10 min Each in at three axis OR (i) Sinusoidal vibration 10-150 Hz Sweep 2g/ 5 min in each of the three mutually perpendicular direction. (ii) 300 Bump 40g, 6 milliseconds in one axis only	Fully wired card cages, power amplifiers, power supplies, modules, sub assemblies etc. - do -
3	Burn-in	168 Hrs at room temperature under maximum load OR 48 Hrs at full load at +55° C	Complete system set - do -

(Refers to Para 3.4 & 3.6)

IN MAINTENANCE PHILOSOPHY

1. **Onboard Repair/Maintenance.** Onboard repair shall be restricted to replacement at PCB/ Modules (LRU) level. Ships staff as part of 1st line maintenance would also undertake routine maintenance and serviceability check/terminal performance checks.
2. **Second Level Repair/Maintenance.** Second level repair/maintenance of the system would normally be undertaken by shore-based unit (Dockyard) and will be restored to when: -
 - (a) Second level maintenance involving higher skills and /or elaborate set ups are required.
 - (b) Periodic parameter checks and record of performance is to be undertaken as part of STW/HATs or after major components/ sub units of the system have been replaced.
3. **Third/Fourth Level Repair/Maintenance.** Third/fourth level repair/maintenance would include following: -
 - (a) PCB/module repairs down to component level.
 - (b) Repairs/overhaul/refurbishing and testing of major assemblies/ complete system.
 - (c) Setting to work, testing and tuning onboard ships after major repairs/overhauls.
 - (d) Repair and calibration of BITE and other test equipment.

